SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – Descriptive Written Exam (5 Questions)

Course Code:	DSA06	**Course Title:**	Data Handling and Visualization
CO Assessed:	CO1 – Precision (BL2)	**Assessment:**	Assessment Tool 1 – Descriptive Written Exam
Weightage:	40% of CIA	**Total Marks:**	50(5 Questions × 10 Marks)
CO1 Statement:	*Understand the fundamental concepts, principles, and techniques of data visualization.(BL2)*		
Student Name:	Gadwal Mohammad Muzzamil	**Reg. No.:**	192424279
Section / Batch:	AI &DS 2024/2028	**Date:**	11/07/2026

Code of Conduct

I ______________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: ______________________

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Data Mapping & Aesthetics	Mapping data types (categorical, ordinal, quantitative) to aesthetics like position, size, shape, and color; role of scales	1	10
Coordinate Systems & Axes	Cartesian vs polar systems; role of axes; use of nonlinear scales (logarithmic)	2	10
Scales in Visualization	Types of scales (linear, logarithmic, ordinal, color); impact on perception and misinterpretation	3	10
Use of Color in Visualization	Color for distinction, representation, highlighting; types of color scales	4	10
Designing Effective Color Palettes	Principles like contrast, harmony, accessibility, and context in palette design	5	10
	GRAND TOTAL:		**50 Marks**

Annexure A – Descriptive Written Exam

1.Explain the concept of mapping data onto aesthetics in data visualization. Discuss how different types of data (categorical, ordinal, and quantitative) are represented using visual aesthetics such as position, size, shape, and color. Illustrate your answer with suitable examples.

2. Describe the role of scales in data visualization. Explain how scales map data values onto visual aesthetics and discuss different types of scales, including linear, logarithmic, ordinal, and color scales. Analyze the impact of inappropriate scaling on data interpretation.

3. Discuss the importance of coordinate systems and axes in visualizing data. Compare Cartesian and polar coordinate systems, highlighting their characteristics, advantages, limitations, and suitable application scenarios with examples.

4. Explain the significance of nonlinear axes in data visualization. Discuss logarithmic and other nonlinear scales, their advantages in representing wide-ranging datasets, and situations where they are more effective than linear axes. Support your explanation with examples.

5. Describe the role of color in data visualization. Differentiate between using color to distinguish categories, represent data values, and highlight important information. Discuss the principles for choosing effective color palettes, including contrast, accessibility, and visual harmony, and explain how poor color choices can affect data interpretation.

Reg: 192424279.
G.MD.Muzammil.

Assessment

Course: Data handling & visualisation
Course code: DSA0607

1) The concept of mapping data onto aesthetics:

It mean giving a unstructed data into representing through visuals.

The main aesthetics are four:

* Shape:
* Size:
* Colour:
* Position:

=> These helps to present data in a meaningful visual form for better understanding

Types of Data: Categorial Data, ordinal & Quantative data.

=> Categorial data represents different groups of categories

• Example: Gender, Department, product type

=> ordinal data: Represents data with a specific order.

• Example: Low, medium, High.

=> Quantative data: Represents numerical data that can be measured

• Example: Age, Salary, sales

Visual aesthetics:

=> Position: It shows the ~~the~~ positions like in a leaderboard the position of players. ~~It~~ shows the exact location of the data on x and y axes. It is the more accurate

Size: Represents the magnitude of data. Larger indicates a higher value

Shape: Used to differentiate Categories using Symbols

Color: Different Colors distinguish categories, while color intensity represent numerical values.

(2) Roles of Scale:

- A scale converts data values into visual elements such as position, color or size.

⇒ It helps represent data accurately and allow easy comparison.

Types of scale:

- Linear scale
- Logorithmic scale
- ordinal scale
- Color scale:

⇒ Linear scale: Equal interval represent equal differences.

Logorithmic scale: used for very large or rapidly increasing value such as for continuous ~~numerical data~~ population or earthquake

Ordinal scale: Used for ordered categories like Poor, Good and Excellent

Color scale: Uses different Colors or color shades to represent data value

Importance of scale:

- Ensures accurate representation of data.
- Make graph easier to understand
- Helps compare values correctly.

Impact of Improper scaling:

- Can exaggerage or hide differences.
- May Confuse viewers.
- Difficult to understand the plots.

Conclusion:

- Scaling convert data values to represent well. Choosing Proper scaling helps in improving of accurate results and easy to understand for the people/users.

5 Role of Colors

- Color is used to improve the appearance and understanding of data.
- It helps to identify categories, represent values and highlight the main information.

Uses of Color:

- Distinguishing Categories: Different colors represent different groups
  - Example: Different colors for product Categories.
- Representing Data values: Color intensity Shows higher or lower values Example: Heatmaps
- Highlighting Information: Bright colors draw attention to important data points

Principles of Good Color selection:

- Contrast:
- Accessibility
- Visual harmony
- Consistency:

Effects of poor color choice:

- Makes charts difficult to read
- Confuses users
- Can result in incorrect interpretation of data.

Conclusions:

× Need to use colors that are easy to distinguish, choose color blind-friendly palettes, using colors that look balanced and pleasant, use the same color for same category. Choosing right color improves clarity to the people.